DHRUV SHARMA

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Expected Graduation: December 2027

EDUCATION

University of California, San Diego — San Diego, CA
M.S. Machine Learning and Data Science — Aug 2026 – Dec 2027
B.S. Computer Engineering, GPA: 3.83/4.0 — Sep 2023 – Dec 2026
Relevant Coursework: Data Structures & Algorithms (DSA), Operating Systems, Computer Architecture, Object-Oriented Programming (OOP), Probability and Statistics, Linear Algebra, Calculus, Data Science, Deep Learning

EXPERIENCE

Primare — Jul 2026 – Present
Software Engineering Intern — San Diego, United States

- Built **Primare OS**, the company's flagship product, as an **MCP gateway** integrating 6 custom and enterprise **MCP** servers, including **GoHighLevel** CRM and Stripe.
- Improved **LLM tool-call** success rate from **76% to 91%** across 10+ production tools by refining API contracts, input schemas, validation logic, and error handling from execution traces, with **NVIDIA NeMo Guardrails**.
- Increased **agent task completion by 20%** by building an **evaluation framework** for multi-step **agentic** workflows, while upholding quality through peer **code review**, **agile** sprints, a **CI suite** across **4 repositories**, and **infrastructure-as-code** (Ansible) deployments monitored by **Prometheus**.
- Reduced **chat latency** through caching and **LLM prefill** optimizations, while cutting permission checks from **14 parallel calls to 1** per page load using single-flight locking and stale-while-revalidate caching, trading freshness for availability during outages.

Knight Lab — Oct 2024 – Present
Graduate Student Researcher — San Diego, United States

- Shipped an open source **full-stack web application** using **agile methodologies**, delivering **natural language processing (NLP)** search, metadata exploration, and collaborative workflows used by **15+ researchers across 600+ studies**. [GitHub]
- Cut study discovery time from **15 minutes to 3 minutes** by writing multi-table **SQL** joins and granular sample-metadata queries over a large dataset, optimizing PostgreSQL queries, lazy-loading large tables, and caching repeated lookups, deployed as a **live server** for the lab.
- Designed a reusable, **model-agnostic agent orchestration** framework connecting **7 LLM backends** with 4 research tools via **JSON-schema** tool arguments and implementing automatic fallback mechanisms for **reliable** multi-step execution.

WeInvest — Feb 2026 – Jun 2026
Software Engineering Intern — Remote, Singapore

- Built a Python **financial research platform** orchestrating **44,000+ backtests** across 78 signal-asset combinations through a nightly **ETL pipeline** scheduled with **cron**.
- Reduced model discovery time from **3 days to 6 hours** by engineering an 8-phase **statistical validation pipeline** with hypothesis testing and adaptive sampling, enforced through **automated integration tests**.
- Built an **LLM-driven autoresearch loop** that generates search boundaries to maximize the **Sharpe ratio** across automated backtest experiments, with run logs recording the model version for every run.

Q-Lab — Feb 2025 – Jun 2025
AI Research Intern — San Diego, United States

- Built a scalable **multi-agent simulation system** supporting 100+ concurrent interactions by integrating neural networks with Unreal Engine **3D visualization** environments and designing software components for **distributed and parallel** simulation workloads.
- Automated large-scale environment generation by integrating UnrealCV with Shap-E to generate **3D** assets and environments, developing **200+ validated** social-interaction animations and reducing manual setup.

RESEARCH & PUBLICATIONS

SimWorld: An Open-ended Simulator for Agents in Physical and Social Worlds
Accepted as NeurIPS 2025 Spotlight, as part of Q-Lab [Paper]

Windowed Principal Coordinate Analysis — *Knight Lab, UC San Diego*

- First author on **windowed Principal Coordinate Analysis** (a dimensionality-reduction algorithm for biological datasets), implementing a streaming, **out-of-core** version in **C++** that moves data between disk and RAM with **mmap**, with my own **multi-GPU** implementation verified by **unit tests**, achieving machine-precision convergence at **6**× lower peak RAM than the industry-standard; manuscript in preparation.

SKILLS & INTERESTS

Languages: Python, C/C++, CUDA, Bash, SQL, Java, JavaScript, TypeScript
Cloud Infrastructure: AWS, Google Cloud, Kubernetes, Docker, Ansible, GitLab CI (CI/CD), Git/GitHub, Unix/Linux
Frameworks & Tools: React, Node.js, FastAPI, PyTorch, PostgreSQL/pgvector, Prometheus, LangChain, MCP, Claude Code
Interests: Chess, Running, Squash, Tennis, Violin